

Chapter 1

Introduction

Project Title: LSTM-Based Navigation Sensor Error Minimization Using Deep Learning

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1.1 Introduction to Navigation Systems

Navigation systems are essential in modern aerospace, defence, autonomous vehicles, missiles, aircraft, unmanned aerial vehicles (UAVs), and marine platforms. Their primary function is to determine the position, velocity, and orientation (attitude) of a moving object. Accurate navigation is critical for maintaining trajectory, ensuring mission success, and enabling safe autonomous operation. Modern navigation systems integrate multiple sensors to provide continuous estimation of motion even in environments where satellite-based navigation is unavailable.

1.2 Inertial Navigation System (INS)

An Inertial Navigation System (INS) estimates the attitude and motion of a vehicle using inertial measurements obtained from accelerometers and gyroscopes. Magnetometers are commonly incorporated to improve heading estimation. Unlike GPS, INS does not rely on external signals and therefore continues to operate in GPS-denied environments. However, the integration of sensor measurements causes errors to accumulate over time, resulting in drift and reduced navigation accuracy.

1.3 Navigation Sensors

Sensor	Purpose	Typical Errors
Accelerometer	Measures linear acceleration	Bias, noise, scale factor
Gyroscope	Measures angular velocity	Bias drift, random walk
Magnetometer	Measures Earth's magnetic field for heading	Magnetic interference, calibration error

1.4 Sources of Navigation Errors

The major sources of error in navigation sensor data include:

- Sensor bias: Constant offset in sensor measurements.
- Drift: Error accumulation over time, particularly in gyroscopes.
- Random noise: High-frequency fluctuations caused by electronic components.

- Scale-factor error: Incorrect proportional relationship between measured and true values.
- Temperature effects: Variations in sensor behaviour due to environmental conditions.
- Magnetic disturbances: External magnetic fields affecting heading estimation.

1.5 Motivation for AI-Based Error Minimization

Traditional navigation algorithms such as the Kalman Filter require accurate mathematical models and careful parameter tuning. In highly nonlinear and dynamic environments, modelling uncertainties may limit estimation accuracy. Deep learning provides a data-driven alternative capable of learning complex temporal relationships directly from historical sensor observations. Among deep learning models, Long Short-Term Memory (LSTM) networks are particularly effective for sequential time-series data because they can capture long-term temporal dependencies while mitigating the vanishing gradient problem.

1.6 Why LSTM?

LSTM networks are designed to process sequential information. Since navigation sensor measurements are collected continuously over time, the current attitude depends not only on the present sensor values but also on previous observations. LSTM maintains an internal memory through gated mechanisms, enabling it to learn temporal behaviour associated with sensor drift, noise, and nonlinear motion. This makes LSTM a suitable choice for navigation sensor error minimization.

1.7 Objectives of the Present Work

The objectives of this work are:

- Develop an LSTM-based model for navigation attitude estimation.
- Minimize roll (ϕ), pitch (θ), and yaw (ψ) estimation errors.
- Evaluate the proposed model using six independent navigation datasets.
- Assess performance using RMSE, MAE, and R^2 metrics.
- Analyse model generalization under varying operating conditions.
- Identify limitations and propose future improvements.

1.8 Chapter Summary

This chapter introduced the importance of navigation systems, inertial sensors, and the challenges associated with sensor errors. The motivation for adopting LSTM-based deep learning for attitude error minimization was discussed along with the objectives of the

research. The following chapter presents the proposed methodology, including data preprocessing, sequence generation, LSTM architecture, and incremental learning.

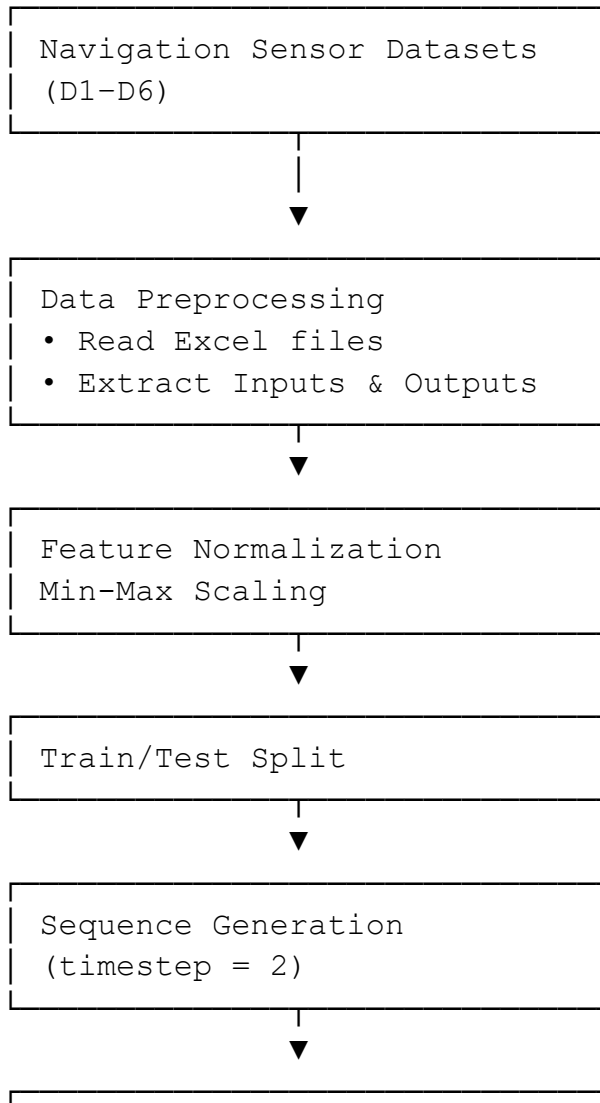
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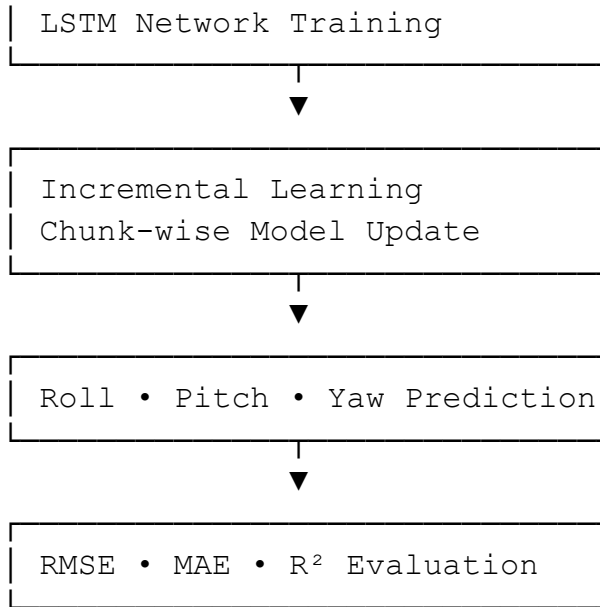
Proposed Methodology

2.1 Overview

This chapter describes the complete methodology adopted for navigation sensor error minimization using a Long Short-Term Memory (LSTM) network. The workflow consists of data acquisition, preprocessing, feature normalization, sequence generation, LSTM model training, incremental learning, prediction, and performance evaluation.

2.2 Overall Workflow





2.3 Dataset Description

Six navigation datasets (D1–D6) were used. Each dataset contains 12 columns. The first nine columns are sensor measurements and the last three columns are the target attitude angles.

Feature Type	Variables	Description
Inputs	ax, ay, az	Linear acceleration
Inputs	p, q, r	Angular velocity
Inputs	mx, my, mz	Magnetic field
Outputs	φ , θ , ψ	Roll, Pitch, Yaw

2.4 Data Preprocessing

Each Excel dataset is loaded into Python using Pandas. The first nine columns are extracted as input features (X), while the last three columns are extracted as target outputs (Y). Missing values are checked before model training. Input features are reshaped according to the LSTM input format.

2.5 Feature Normalization

Min-Max normalization is applied to the input features to ensure that all variables are within a common numerical range. This improves convergence and stabilizes gradient updates during training.

Min-Max Scaling Formula:

$$X_{\text{normalized}} = (X - X_{\text{min}}) / (X_{\text{max}} - X_{\text{min}})$$

2.6 Train-Test Split

The normalized dataset is divided into training and testing subsets. The training data is used to learn the temporal mapping between sensor measurements and attitude angles, while the testing data evaluates the model's generalization performance.

2.7 Sequence Generation

The LSTM requires a three-dimensional input tensor of shape (samples, timesteps, features). In this work, a timestep of 2 is used. Consecutive sensor samples are grouped into sequences before being supplied to the network.

Input Shape = (Number of Sequences, Timestep, Number of Features)

Example: (5000, 2, 9)

2.8 LSTM Architecture

The implemented network consists of:

- Input Layer
- LSTM Layer (50 units)
- Dropout Layer (0.25)
- LSTM Layer (20 units)
- Dense Output Layer (3 neurons with linear activation)

The model uses Mean Squared Error (MSE) as the loss function and RMSprop optimizer.

2.9 Incremental Learning

Instead of testing the model only once, the test data is divided into chunks. After predicting each chunk, the model is retrained using that chunk before proceeding to the next. This simulates online adaptation where the network continuously learns from newly arriving navigation data.

2.10 Performance Metrics

The following metrics are used:

- RMSE (Root Mean Square Error): Measures average prediction error.
- MAE (Mean Absolute Error): Measures average absolute deviation.
- R² Score: Indicates how well the model explains the variation in the target values.

2.11 Hyperparameters

Parameter	Value
Timestep	2
Input Features	9
Output Features	3
LSTM Units	50, 20
Dropout	0.25
Batch Size	50
Epochs	20
Loss Function	Mean Squared Error
Optimizer	RMSprop

2.12 Chapter Summary

This chapter described the complete processing pipeline from raw navigation sensor data to attitude prediction. The preprocessing steps, sequence generation, LSTM architecture, incremental learning strategy, and evaluation metrics establish the foundation for the experimental analysis presented in the next chapter.

Chapter 3

Experimental Results and Analysis

This chapter presents the experimental evaluation of the proposed LSTM-based navigation sensor error minimization model on six datasets (D1–D6). Performance is evaluated using RMSE, MAE and R^2 score. For each dataset, the predicted Roll (ϕ), Pitch (θ) and Yaw (ψ) are compared with the reference values to assess the capability of the model to learn temporal navigation dynamics.

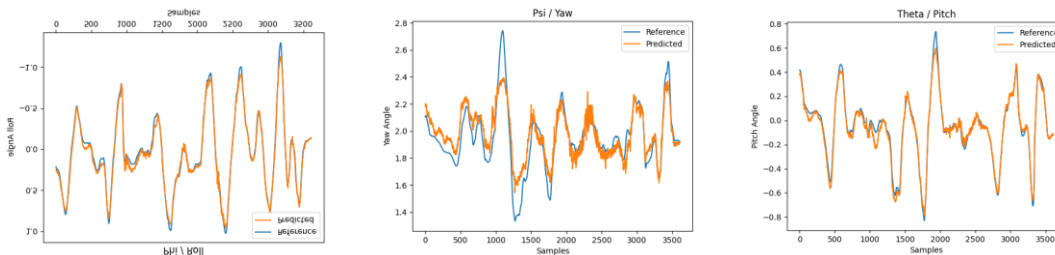
3.1 Performance Comparison

Dataset	Roll RMSE	Pitch RMSE	Yaw RMSE	Roll R^2	Pitch R^2	Yaw R^2	Overall
D1	0.0472	0.0383	0.1090	0.9894	0.9801	0.7760	Excellent
D2	0.0699	0.0423	0.1010	0.9643	0.9389	0.6921	Very Good
D3	0.0272	0.0455	0.1268	0.9928	0.9653	-0.1420	Yaw degraded
D4	0.3807	0.0572	0.4171	-2.055	-2.842	-0.8348	Failed
D5	0.9956	0.0561	0.3185	-2.625	-0.542	0.2318	Failed
D6	1.3675	0.5203	1.0387	0.4610	0.6256	0.4670	Moderate

3.2 Dataset D1

The model achieved excellent agreement between predicted and reference attitude values. Roll and Pitch closely followed the reference trajectory with very small deviations. Yaw exhibited slightly higher error but the overall trend was captured accurately. This dataset demonstrates that the proposed LSTM can effectively learn temporal navigation dynamics under familiar operating conditions.

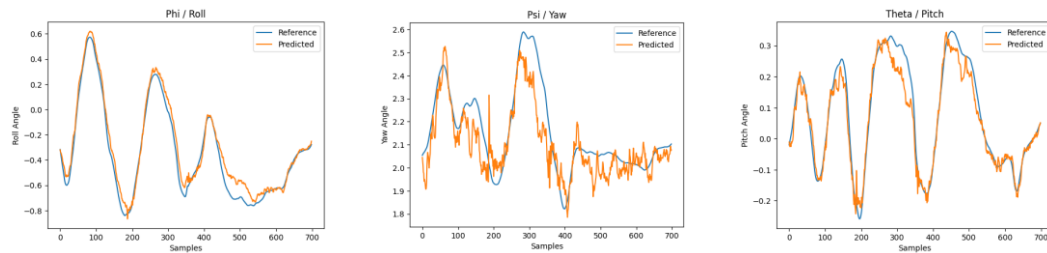
Figures: Insert the Roll, Pitch, Yaw prediction plots and metrics screenshot corresponding to Dataset D1 below.



3.3 Dataset D2

The model maintained high prediction accuracy with only a slight reduction compared to D1. Roll and Pitch remained highly accurate while Yaw showed moderate degradation. Overall, the model generalized well to this dataset.

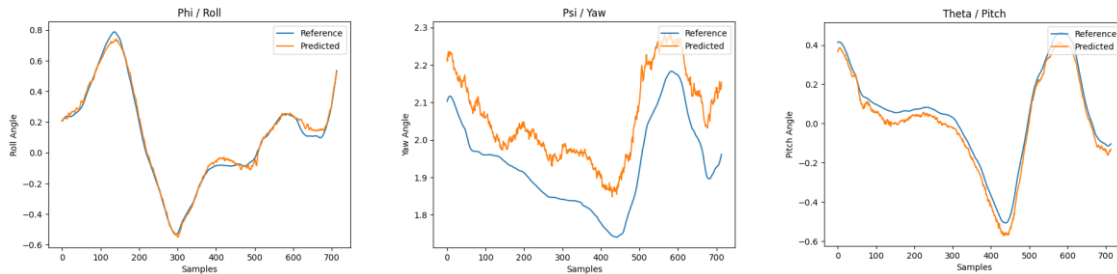
Figures: Insert the Roll, Pitch, Yaw prediction plots and metrics screenshot corresponding to Dataset D2 below.



3.4 Dataset D3

Roll and Pitch predictions remained excellent. However, Yaw prediction degraded significantly, resulting in a negative R^2 value. This indicates difficulty in modeling heading behaviour, likely due to drift or different motion characteristics.

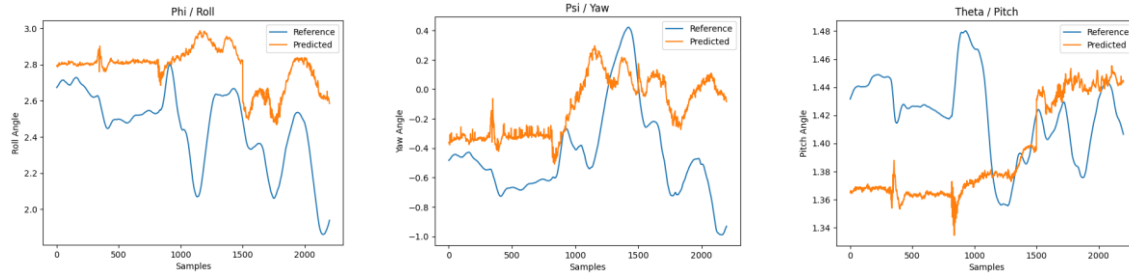
Figures: Insert the Roll, Pitch, Yaw prediction plots and metrics screenshot corresponding to Dataset D3 below.



3.5 Dataset D4

The model failed to generalize on this dataset. Large offsets between predicted and reference values produced negative R^2 scores for all outputs, indicating a substantial distribution shift from the training data.

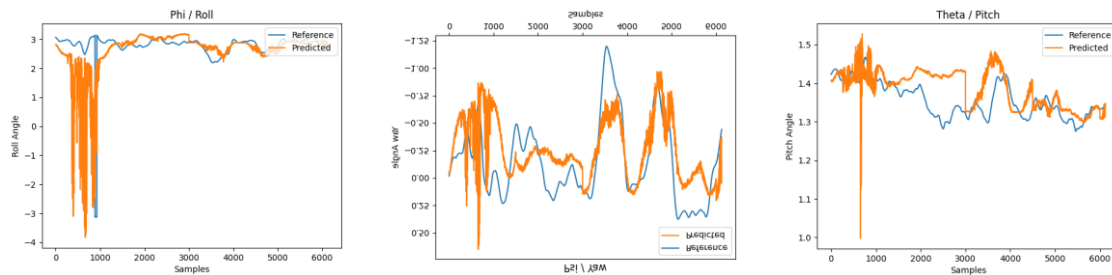
Figures: Insert the Roll, Pitch, Yaw prediction plots and metrics screenshot corresponding to Dataset D4 below.



3.6 Dataset D5

Prediction quality deteriorated further, particularly for Roll. Although Yaw showed slight improvement over D4, overall performance remained poor, suggesting different operating conditions or sensor characteristics.

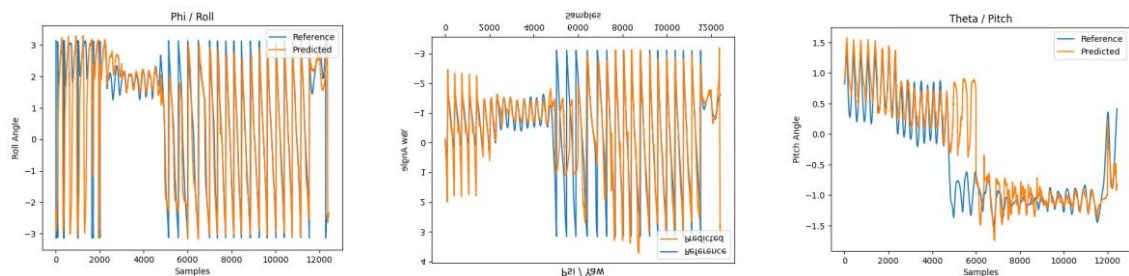
Figures: Insert the Roll, Pitch, Yaw prediction plots and metrics screenshot corresponding to Dataset D5 below.



3.7 Dataset D6

The model successfully captured the oscillatory behaviour of the signals but underestimated the amplitude and exhibited phase differences. Consequently, moderate R^2 values were obtained, indicating partial generalization.

Figures: Insert the Roll, Pitch, Yaw prediction plots and metrics screenshot corresponding to Dataset D6 below.



3.8 Overall Discussion

Among the six datasets, D1 and D2 produced the best results, indicating that the LSTM architecture successfully learned temporal relationships under operating conditions similar to the training data. D3 demonstrated that yaw estimation remains challenging due to accumulated heading errors and sensor drift. D4 and D5 highlighted a lack of cross-dataset generalization, suggesting significant distribution shift. D6 showed that the model learned the overall waveform but struggled with amplitude and phase accuracy. These findings indicate that the proposed model serves as a strong baseline but requires improved generalization through multi-dataset training, larger sequence lengths, and enhanced architectures.

3.9 Key Findings

- Roll estimation consistently outperformed Yaw estimation.
- Yaw prediction is more sensitive to sensor drift and magnetic disturbances.
- Model performance depends strongly on similarity between training and testing distributions.
- Incremental learning helps adaptation but does not fully overcome distribution shift.
- D1 and D2 validate the effectiveness of the proposed LSTM approach.

Chapter 4

Conclusion, Limitations and Future Scope

4.1 Summary of the Research

This work investigated the application of Long Short-Term Memory (LSTM) networks for navigation sensor error minimization using six independent navigation datasets. The proposed approach utilized nine sensor measurements (accelerometer, gyroscope and magnetometer) as inputs to estimate the three attitude angles: Roll (ϕ), Pitch (θ) and Yaw (ψ). Data preprocessing, Min-Max normalization, sequence generation and incremental learning were incorporated to improve the learning capability of the network.

4.2 Major Findings

- Datasets D1 and D2 achieved excellent prediction accuracy with high R^2 scores, demonstrating that the LSTM effectively learned temporal relationships in the navigation data.
- Dataset D3 produced accurate Roll and Pitch predictions, while Yaw estimation degraded considerably, indicating that heading estimation remains more challenging.
- Datasets D4 and D5 showed poor generalization, suggesting that the testing data distribution differed significantly from the training data.
- Dataset D6 demonstrated moderate performance. Although the model captured the oscillatory trend, amplitude and phase mismatches reduced the prediction accuracy.
- Roll and Pitch estimation consistently outperformed Yaw estimation across most datasets.

4.3 Limitations of the Present Work

- Model training was primarily based on a single dataset, limiting cross-dataset generalization.
- A timestep of 2 may be insufficient to capture long-term temporal dependencies in navigation signals.
- The optimizer and hyperparameters were not extensively tuned.
- Yaw estimation remains sensitive to sensor drift and magnetic disturbances.
- Incremental learning improves adaptation but may lead to catastrophic forgetting if not carefully controlled.

4.4 Future Scope

- Increase the sequence length (timestep) to capture richer temporal information.
- Train a unified model using multiple datasets to improve robustness.
- Perform systematic hyperparameter optimization for learning rate, batch size, epochs and LSTM units.
- Evaluate Bidirectional LSTM, GRU, Attention-LSTM and Transformer-based architectures.
- Compare the proposed method with Kalman Filter, Extended Kalman Filter and hybrid Kalman–LSTM approaches.
- Implement cross-validation across the six datasets to better assess generalization.
- Investigate domain adaptation techniques to reduce performance degradation under unseen operating conditions.
- Deploy the trained model on embedded hardware for real-time navigation applications.

4.5 Contributions of the Work

Contribution	Outcome
Navigation sensor preprocessing	Prepared six datasets for sequential learning
LSTM-based attitude estimation	Estimated Roll, Pitch and Yaw from sensor data
Incremental learning	Adapted model using chunk-wise updates
Experimental evaluation	Compared performance across six datasets
Performance analysis	Identified strengths, limitations and generalization issues

4.6 Conclusion

The proposed LSTM-based navigation sensor error minimization framework demonstrates that deep learning can accurately estimate attitude parameters from sequential inertial sensor data. The experimental results confirm excellent performance for datasets with dynamics similar to the training distribution, while also revealing the need for improved robustness when operating under different conditions. The findings establish a strong baseline for future research on AI-assisted navigation systems. Incorporating multi-dataset training, longer temporal windows, advanced neural network architectures and hybrid filtering techniques is expected to significantly improve generalization and real-world deployment capability.

4.7 Overall Report Summary

This project successfully demonstrated the feasibility of using LSTM networks for navigation sensor error minimization. Through comprehensive experimentation on six datasets and evaluation using RMSE, MAE and R^2 metrics, the study highlighted both the strengths and limitations of the current approach. The methodology, experimental analysis and future recommendations presented in this report provide a solid foundation for extending the work towards robust AI-driven navigation solutions for defence and autonomous systems.